Profile Summary (Total time: 19.911 s)

Generated 16-May-2026 07:26:21 using performance time.

Function Name	Calls	Total Time (s)	Self Time* (s)	Total Time Plot (self time / total time)

driver	1	19.911	0.025	
driver>solve_kinematic_fmincon	1	19.886	0.001	
driver>run_once	1	19.831	0.002	
fmincon	1	19.829	0.019	
barrier	1	19.633	0.153	
driver>nl_con	369	19.141	2.159	
constrfungradEvaluator	368	19.097	0.005	
driver>@(x)nl_con(x,prob)	368	19.092	0.004	
OptimFunctions>OptimFunctions.constraints	367	19.089	0.082	
RigidBodyTree>RigidBodyTree.forwardKinematics	36900	10.143	1.151	
rigidBodyTree>rigidBodyTree.geometricJacobian	18450	8.083	0.039	
RigidBodyTree>RigidBodyTree.geometricJacobian	18450	8.044	1.160	
rigidBodyJoint>rigidBodyJoint.transformBodyToParent	369000	7.338	0.651	
rigidBodyJoint>rigidBodyJoint.jointTransform	369000	6.687	0.705	
rigidBodyTree>rigidBodyTree.getTransform	18450	6.128	0.043	
RigidBodyTree>RigidBodyTree.getTransform	18450	6.085	0.208	
axang2tform	221400	4.409	0.331	
axang2tform	221400	3.476	0.324	
axang2rotm	221400	2.244	1.347	
rigidBodyJoint>rigidBodyJoint.get.Type	848700	1.939	0.828	
SorosimContactPair>SorosimContactPair.solveContact	88560	1.656	0.630	
CharacterVector>CharacterVector.getVector	1151280	1.489	1.489	
RigidBodyTree>RigidBodyTree.parentFrameToParentBodyTform	369000	1.302	0.872	
RigidBodyTree>RigidBodyTree.validateInputFrameName	36900	0.979	0.054	
RigidBodyTree>RigidBodyTree.findBodyIndexByName	36900	0.925	0.297	
rotm2tform	221400	0.908	0.908	
normalizeRows	221400	0.897	0.897	
idcol_solve_mex (MEX-file)	88560	0.789	0.789	
RigidBody>RigidBody.get.Name	302580	0.712	0.335	
rigidBodyJoint>rigidBodyJoint.get.JointAxis	221400	0.695	0.222	
validateNumericMatrix	221400	0.602	0.602	
SorosimContactPair>SorosimContactPair.get_relative	177120	0.484	0.320	
FrameTree>FrameTree.transformUsingFrameID	405900	0.474	0.474	
SorosimContactBody>SorosimContactBody.get_gradH	88560	0.360	0.188	
repmat	36900	0.324	0.058	
ginv	354240	0.322	0.322	
tformToSpatialXform	73800	0.317	0.242	
rigidBodyJoint>rigidBodyJoint.get.MotionSubspace	73800	0.292	0.129	
repmat	36900	0.265	0.214	
dinamico_Adjoint	177120	0.215	0.215	
RigidBodyTree>RigidBodyTree.validateConfiguration	36900	0.175	0.175	
shape_core_mex (MEX-file)	88560	0.172	0.172	
optim\private\computeTrialStep	383	0.169	0.035	
dinamico_tilde	177120	0.136	0.136	

RigidBodyTree>RigidBodyTree.transformToBaseFrame	36900	0.135	0.091	
OptimFunctions>OptimFunctions.constraintFirstEval	1	0.116	0.004	
RigidBody>RigidBody.get.Joint	147600	0.108	0.108	
optim\private\formAndFactorKKTmatrix	101	0.074	0.069	
skew	73800	0.074	0.074	
RigidBodyTree>RigidBodyTree.get.BaseName	18450	0.067	0.024	
OptimFunctions>OptimFunctions.objective	367	0.055	0.032	
repmat>checkSizesType	73800	0.051	0.051	
varsize	73800	0.042	0.042	
…idBodyTree>RigidBodyTree.assertUpperBoundOnVelocityNumber	18450	0.037	0.024	
optim\private\nlpStopTest	102	0.030	0.027	
OptimFunctions>OptimFunctions.computeGradAndJac	102	0.029	0.009	
optim\private\tangentialStep	68	0.028	0.005	
RigidBodyTree>RigidBodyTree.assertUpperBoundOnNumBodies	36900	0.028	0.028	
OptimFunctions>OptimFunctions.setup	1	0.023	0.005	
objfungradEvaluator	368	0.023	0.004	
OptimFunctions>reshapeAndCastOutputs	2505	0.021	0.021	
optim\private\projConjGrad	68	0.020	0.013	
driver>@(x)obj_fun(x,prob)	368	0.018	0.003	
OptimFunctions>OptimFunctions.constraintgrad	101	0.015	0.006	
driver>obj_fun	368	0.015	0.015	
optim\private\formJacobian	102	0.015	0.009	
optim\private\solveAugSystem	505	0.015	0.005	
optim\private\backsolveSys	609	0.015	0.015	
optim\private\formAndFactorAugMatrix	35	0.014	0.013	
optim\private\normalStep	68	0.013	0.002	
RigidBodyTree>RigidBodyTree.upperBoundOfPositionNumber	18450	0.013	0.013	
optim\private\updatePenaltyParam	365	0.012	0.009	
spdiags	440	0.011	0.010	
optim\private\computeHessian	102	0.009	0.009	
optim\private\formConstraints	368	0.009	0.009	
…tionScaling>optimizationFunctionScaling.initializeScaling	1	0.009	0.006	
…timizationFunctionEvaluator.optimizationFunctionEvaluator	1	0.008	0.002	
…caleConstrGrad(self.cIneq.',self.cEq.',cineqGrad,ceqGrad)	102	0.008	0.002	
callAllOptimOutputFcns	104	0.008	0.002	
OptimFunctions>OptimFunctions.objectiveFirstEval	1	0.007	0.004	
optim\private\xFixedAndBounds	383	0.007	0.007	
OptimFunctions>OptimFunctions.optimfcncheck	1	0.007	0.004	
optim\private\normalCauchyStep	68	0.007	0.003	
optim\private\solveKKTsystem	104	0.007	0.002	
OptimFunctions>OptimFunctions.setFiniteDifferenceFlags	1	0.007	0.005	
optim\private\barrierTestAndUpdate	101	0.006	0.006	
saveFminconSolution	107	0.006	0.005	
optim\private\acceptanceTest	383	0.006	0.006	

…)saveFminconSolution(x1,x2,x3,trackerID,TolCon,verbosity)	104	0.006	0.001	
OptimFunctions>OptimFunctions.OptimFunctions	1	0.006	0.006	
parseOptions	1	0.006	0.001	
optimizationFunctionScaling>scaleConstrGrad	102	0.006	0.006	
optim\private\leastSquaresLagrangeMults	38	0.006	0.002	
…ng>@(cIneq,cEq)scaleConstr(self.cIneq,self.cEq,cIneq,cEq)	368	0.005	0.003	
getIpOptions	1	0.005	0.003	
prepareOptionsForSolver	1	0.005	0.000	
optim\private\hessTimesVector	751	0.005	0.005	
OptimFunctions>OptimFunctions.objgrad	101	0.005	0.004	
createExitMsg	2	0.005	0.003	
SolverOptions>SolverOptions.extractOptionsStructure	1	0.004	0.000	
optim\private\normalNewtonStep	68	0.004	0.002	
Fmincon>Fmincon.extractCustomOptionsStructure	1	0.004	0.001	
OptimFunctions>OptimFunctions.setOptions	1	0.004	0.003	
…ltiAlgorithm>MultiAlgorithm.extractCustomOptionsStructure	1	0.003	0.002	
optimizationFunctionScaling>@(x)scaleObjOrGrad(objScale,x)	470	0.003	0.002	
checkbounds	1	0.003	0.001	
isoptimargdbl	3	0.003	0.003	
formLambdaStruct	1	0.002	0.002	
optimget	48	0.002	0.001	
optim\private\truncateTangStep	68	0.002	0.001	
createCellArrayOfFunctions	1	0.002	0.001	
optimizationFunctionScaling>scaleConstr	368	0.002	0.002	
fmincon>addSolutionTracker	1	0.002	0.001	
setSizes	1	0.002	0.002	
optim\private\fractionToBoundaryScaled	171	0.002	0.002	
optim\private\backtrack	196	0.001	0.001	
optim\private\fractionToBoundary	103	0.001	0.001	
unique	1	0.001	0.001	
optimfcnchk	2	0.001	0.001	
optim\private\accStepTRupdate	101	0.001	0.001	
OptimFunctions>OptimFunctions.setX0AndBounds	1	0.001	0.001	
replace	1	0.001	0.001	
createCellArrayOfFunctions>isFcn	1	0.001	0.000	
optimizationFunctionScaling>scaleObjOrGrad	470	0.001	0.001	
Fmincon>Fmincon.replaceSpecialStrings	1	0.001	0.001	
optimget>optimgetfast	39	0.001	0.001	
fmincon>checkSolutionTracker	1	0.001	0.001	
optim\private\fractionToBoundaryHonorBounds	103	0.001	0.001	
optim\private\fractionToBoundaryTangential	68	0.001	0.001	
ObjectiveSenseManager>ObjectiveSenseManager.setup	1	0.001	0.001	
optimlib\private\classifyBoundsOnVars	1	0.001	0.001	
spdiags>isNonnegativeInt	880	0.001	0.001	

setHiddenFminconOptions	1	0.001	0.001	
OptimFunctions>OptimFunctions.setScale	1	0.001	0.001	
getNumericOrStringFieldValue	4	0.001	0.001	
spdiags>isInt	440	0.001	0.001	
createExitMsg>msgArgs2Str	3	0.001	0.001	
OptimFunctions>OptimFunctions.setObjectiveEvaluator	1	0.001	0.001	
strip	1	0.001	0.001	
OptimFunctions>OptimFunctions.setConstraintEvaluator	1	0.001	0.001	
fcnchk	1	0.001	0.001	
validateFinDiffRelStep	1	0.001	0.001	
SolverOptions>SolverOptions.convertForSolver	1	0.001	0.001	
unique>uniqueR2012a	1	0.000	0.000	
MultiAlgorithm>MultiAlgorithm.get.Algorithm	1	0.000	0.000	
optim\private\dampingProcedure	101	0.000	0.000	
SolverOptions>SolverOptions.isSetByUser	4	0.000	0.000	
checkForOptimPlot	1	0.000	0.000	
ObjectiveSenseManager>@(x)full(x)	102	0.000	0.000	
Fmincon>Fmincon.get.FinDiffRelStep	1	0.000	0.000	
strfun\private\isTextStrict	2	0.000	0.000	
checkoptionsize	1	0.000	0.000	
replaceFinDiffRelStepString	1	0.000	0.000	
validateUseParallel	1	0.000	0.000	
fmincon>@()saveFminconSolution(trackerID)	1	0.000	0.000	
SolverOptions>SolverOptions.getOptionsStore	1	0.000	0.000	
driver>@()cellfun(@(s)warning(s.state,s.identifier),saved)	1	0.000	0.000	
strfun\private\convertStringToOriginalTextType	2	0.000	0.000	
OptimFunctions>functionHandleOrEmpty	11	0.000	0.000	
…g>optimizationFunctionScaling.optimizationFunctionScaling	1	0.000	0.000	
driver>pack_solution	1	0.000	0.000	
SolverOptions>SolverOptions.mapOptionsForSolver	1	0.000	0.000	
driver>pick_bestfeasible	1	0.000	0.000	
driver>@(s)warning(s.state,s.identifier)	3	0.000	0.000	

***Self time** is the time spent in a function excluding any time spent in child functions. The time includes any overhead time resulting from the profiling process.